

Symmetry-Protected Dynamics in Open Quantum Systems

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Abstract

I. INTRODUCTION

THE theories of quantum and classical probability both have epistemic issues. Classical probability debates over frequentist versus Bayesian interpretations. However, quantum probability theory has a more ontological issue, regarding the meaning, or lack thereof, of the theory as a whole. In fact, quantum logic essentially differs fundamentally from classical logic through the failure of distributivity on a probabilistic lattice. Furthermore, in 1927, John Von Neumann pointed out that quantum randomness was fundamentally different from classical randomness [1]. All of this makes quantum probability alluring and using what we know to help improve our understanding of what it is. This different framework presents a very natural setting to examine the impacts of symmetry on probabilistic quantum dynamical processes. As Eugene Wigner emphasized in his original 1927 work, with the help of representation-theoretic symmetry arguments, much can be derived “almost entirely without calculation” [2].

II. QUANTUM STATES AND OPEN-SYSTEM EVOLUTION

A. Quantum Randomness

For those familiar with measure theory or classical probability theory, there is nothing inherently “random” at all about it. For brevity, for every possible event or set of events that can occur, a probability is assigned to that set, representing the probability of occurring, all else equal. The relationships between these events are described through a Boolean algebra, which expresses information about compatibility of certain events. All these events live in the same space.

Quantum probability theory is inherently different. For each observable $A \in \mathcal{B}(\mathcal{H})$ such that the operator A on Hilbert space $\mathcal{H}$ is self-adjoint¹ $A^* = A$, the spectral theorem assigns a projection-valued measure E_A . What this tells us is that the possible events corresponding to measurements of different values of A are all represented by projection operators.

B. Bayesian Statistics

There is a big difference in representing states as density operators versus vectors. In the case of a discrete system, our density operators ρ are distributions of a discrete state space of possible states to transition to.

So the Lindblad equation is the unconditioned evolution. A trajectory is the conditioned evolution, conditioned on a particular measurement/noise record.

For a discrete-time process, the probability for the system to be in a state after k steps given some probability distribution P on the transition process is the convolution of P by itself k -times:

$$P(x|t = \tau k) = P^{*k}(x)$$

A Bayesian update is one as follows:

$$\rho \mapsto \frac{M_y \rho M_y^*}{\text{Tr}(M_y^* M_y \rho)}, \quad (1)$$

where $P(\rho|y) = dt \text{Tr}(M_y^* M_y \rho)$.

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¹Events are represented by Hermitian operators since these operators have real-valued spectra, which is what we can measure.

III. SYMMETRIES AND CONSERVATION

A. Group Representations

The study of physical systems is entirely based around the analysis of symmetries and patterns and their impacts through phenomenal observable properties. Groups are abstract mathematical concepts, and in order to apply them to our systems, we need a way to “represent” them on our space. We do this through a structure-preserving map known as a homomorphism.

Lie groups are governed by the exponential map, which gives an ODE, which is how the Lindblad arises.

- Representations of groups are called unitary if their images are unitary matrices. Representations of Lie algebras are called unitary if their images are skew-Hermitian matrices

B. Invariant Subspaces and Block Decomposition

When we say that a physical system has a symmetry group G , it should mean that the overall dynamics of the system do not change when the system is transformed according to said G . Specifically, we want to ensure that the energies remain the same under those transformations $g \in G$, which is a condition we enforce with conjugation by U

$$\hat{U}^{-1} \hat{H} \hat{U} \equiv \hat{U}^* \hat{H} \hat{U} = \hat{H} \quad \text{if and only if} \quad [\hat{H}, \hat{U}] = 0 \quad \text{for every } g \in G, \quad (2)$$

where $\hat{U} \equiv \hat{U}(g)$ is the representative of the group-element g and $[\hat{H}, \hat{U}] = 0$ is the commutation relation. This means that if we perform a symmetry transformation on an eigenstate of $\hat{H}$, it is mapped to another state with the same energy:

$$E_{\hat{U}\psi} = \langle \hat{U}\psi, \hat{H}\hat{U}\psi \rangle = \langle \psi, \hat{U}^* \hat{H} \hat{U} \psi \rangle = \langle \psi, \hat{H} \psi \rangle = E_\psi. \quad (3)$$

If two operators commute, it implies they have shared eigenspaces and thus, in a suitable basis, will share all their eigenvectors. In particular, if an operator $\hat{H}$ commutes with a symmetry operator $\hat{U}$, it means, in the correct basis, $\hat{H}$ will have eigenstates corresponding to those of operators in the symmetry group. In other words, if two operators commute, they can be simultaneously diagonalized. This is something we will take as given, but for a proof, see Appendix B.

C. Invariant Subspaces and Irreducible Representations

Now that we have a representation for a symmetry group G on our Hilbert space, $\mathcal{H}$, we can examine how the symmetries operate on different parts, or subspaces. Different symmetries will naturally impact different subspaces. Specifically, we have the ability to decompose $\mathcal{H}$ into *invariant subspaces* $\mathcal{V} \leq \mathcal{H}$. Mathematically, a subspace $\mathcal{V} \leq \mathcal{H}$ is invariant under a group representation $(\hat{U}, \mathcal{H})$ if

$$\hat{U}(g)\psi \in \mathcal{V} \quad \text{for all } g \in G \text{ and } \psi \in \mathcal{V}. \quad (4)$$

That is, no symmetry transformation will ever cause a state to leave the subspace. Upon decomposition of a Hilbert space into invariant subspaces, we write

$$\mathcal{H} = \bigoplus_{\lambda} \mathcal{H}_{\lambda}, \quad (5)$$

where each $\mathcal{H}_{\lambda}$ is an invariant subspace of $\mathcal{H}$ labeled by λ . For example, if we have two Hamiltonians $\hat{H}_{\alpha}$ and $\hat{H}_{\beta}$ that act on orthogonal subspaces $\mathcal{H}_{\alpha}$ and $\mathcal{H}_{\beta}$, then the Hamiltonian on the total space $\mathcal{H} = \mathcal{H}_{\alpha} \oplus \mathcal{H}_{\beta}$ takes the form

$$\hat{H}_{\alpha} \oplus \hat{H}_{\beta} = \begin{bmatrix} \hat{H}_{\alpha} & 0 \\ 0 & \hat{H}_{\beta} \end{bmatrix}, \quad (6)$$

which is a *block-diagonal matrix*. One interesting thing about this is that we can think about operating on these block-matrices as if they are normal matrices, but the matrix components are operators rather than scalars:

$$(\hat{U}_1^* \oplus \hat{U}_2^*)(\hat{H}_1 \oplus \hat{H}_2)(\hat{U}_1 \oplus \hat{U}_2) = \begin{bmatrix} \hat{U}_1^* \hat{H}_1 \hat{U}_1 & 0 \\ 0 & \hat{U}_2^* \hat{H}_2 \hat{U}_2 \end{bmatrix} \quad (7)$$

However, the importance of block matrices should not be understated as they visually depict how the associated linear map respects (or fails to respect) a subspace decomposition [c.f. Claudio rep-theory notes]. For a given dimension, more blocks means more invariance or more structure to respect.

D. Irreducible Representations

We also have a notion of *irreducibility* for representations. Irreducibility gives us the smallest subspaces we can keep independent from each other knowing the symmetry. An *irreducible representation* is a representation of a group that cannot² be broken down into a direct sum of smaller, nontrivial, invariant subspaces. Explicitly, a G -rep $(\hat{U}, \mathcal{H})$ is said to be *irreducible*, or an *irrep*, if the only invariant subspaces of $\mathcal{H}$ are trivial: $\{0\}$ or $\mathcal{H}$ itself.

A key property here is the representation theoretic property of how “commuting” (in the case of observables, compatible!) operators must share eigenstates. This is extremely powerful, as it puts huge constraints on what we know we can see about our system. This is demonstrated in the following theoretical example: consider an unconstrained system, so one with a continuum of values it can take on. If we (breaking symmetry) know that it obeys some special³ symmetry, so one with a discrete spectrum, that system must also have a discrete spectrum, as all eigenstates must similarly be eigenstates of the symmetry. This can be interpreted as a set-theoretic intersection, or minimal cardinality. Since intersecting spectra will only decrease the size, knowing more and more symmetries of our system corresponds exactly to knowing more and more about the possible values it can take on.

E. Non-Unitary Dynamics and the Lindblad Equation

- "On one hand, the presence of symmetries, i.e., operators commuting with the full dynamical Lie algebra L , implies that the given representation of L is not irreducible, that is, the vector space V , where in (2) lives, splits into a number of invariant subspaces each carrying an irreducible representation of the dynamical Lie algebra L ."

- "Transitions from one subspace to the other are forbidden for the dynamics which results in uncontrollability"

- commutation subgroup is the normal subgroup. normality is defined by the action of the group on itself by conjugation

- We are concerned with the space of endomorphisms $\text{End}(V)$, which if $V \cong \mathbb{C}^n$ can be identified with the space of $n \times n$ matrices with complex entries (no other constraint!).

- The space $\text{End}_G(V)$ is an important one and it denotes the space of elements $\phi \in \text{End}_G(V)$ such that $\phi\rho(g) = \rho(g)\phi$. These maps are intertwiners, equivariants, or just maps that are invariant under the group G .

- If we consider a time-parameterized sequence of densities ρ_t , we may look at the comparison of ρ_t to ρ_{t+dt} through $\rho(dt)$, as the system is assumed to be Markovian, meaning that the update in ρ is independent of where we are in time - purely local in time.

We build the lindblad equation in the same way that we build the unitary operator U . The lindblad is formed by essentially extending the Heisenberg equation.

All Lie groups and the dynamics therein are completely determined through the exponential map (through the definition as a Taylor series) of the Lie algebra. Thus, we shall restrict our attention to the Lie algebra, which we shall denote as $\mathfrak{g}$. A good example of this is the Lindbladian itself, which is governed by the exponential map of the Kraus operators. The formal solution to the lindblad equation

$$\dot{\rho}(t) = \mathcal{L}\rho$$

is

$$\rho(t) = \exp(\mathcal{L}t)(\rho(0))$$

A Lindblad equation evolves by a completely positive trace-preserving semigroup $\mathcal{L}$. In the lindblad, if all damping rates $\gamma_i \geq 0$ actually reach 0, then we recover the von Neumann equation

$$\dot{\rho}(t) = -i/\hbar[H, \rho]$$

²i.e., under no change in basis.

³

For a Lindblad equation,

$$\dot{\rho} = \mathcal{L}(\rho) = -i[H, \rho] + \sum_k \left(L_k \rho L_k^* - \frac{1}{2} \{L_k^* L_k, \rho\} \right),$$

the analogous objects are H , the Lindblad operators L_k , and the algebra they generate. A **decoherence-free subspace** $\mathcal{C}$ roughly satisfies the condition that the noise operators act as scalars on it:

$$L_k |\psi_k\rangle = c |\psi_k\rangle \quad \text{for all } |\psi\rangle \in \mathcal{C}.$$

For a noiseless subsystem, the condition is more flexible: The Hilbert space decomposes like

$$\mathcal{H} \cong \bigoplus_J \mathcal{H}_{A,J} \otimes \mathcal{H}_{B,J},$$

and the noise acts only on one factor, say $\mathcal{H}_{B,J}$, while $\mathcal{H}_{A,J}$ stores protected information.

F. Group-Constrained State Evolution

If we initialize a state entirely within an invariant subspace, then it does not leave that subspace, so dynamics of different subspaces do not mix. However, if we initialize a state as a superposition across invariant subspaces $\mathcal{H}_j$, even unitary time evolution will act on each component separately:

$$\phi(t) = \sum_j e^{-i\hat{H}_j t/\hbar} \phi_j.$$

Here, $\hat{H}_j = P_j \hat{H} P_j$ is the restriction of $\hat{H}$ to the subspace $\mathcal{H}_j$

IV. SYSTEMS

Quantum systems are inherently described in terms of their energies.

A. Symmetry Group Commutation

In the presence of symmetries, i.e., operators commuting with full Lie algebra $\mathfrak{g}$ means that the representation of $\mathfrak{g}$ is not irreducible. If the Hamiltonian $\hat{H}$ commutes with a non-trivial set of symmetry operations G (commutes with the action of G), the Hilbert Space for the system decomposes into a number of invariant subspaces as $\mathcal{H} = \bigoplus_\lambda \mathcal{H}_\lambda$, where each carries an irreducible representation of the Lie algebra $\mathfrak{g}$. Thus, Each $\mathcal{H}_\lambda$ is an “invariant subspace” under $\hat{H}$. That is, when we restrict $\hat{H}$ to operating on any one of the subspaces $\hat{H}|_{\mathcal{H}_j}$, the action of $\hat{H}|_{\mathcal{H}_j}$ is irreducible in the sense that there is no further (smaller) nontrivial subspace of $\mathcal{H}_j$ that is invariant under $\hat{H}$.

The operators that commute with G (the commutant, or centralizer algebra) consists of all block diagonal matrices on these subspaces

$\text{End}_G(\mathcal{H}) = \{\text{all block-diagonal matrices}\}$ Constructing a general symmetric $\hat{H}$ reduces to assigning a Hermitian matrix on each irrep block/ invariant subspace Why hermitian? When two operators commute, the eigenstates of one operator can be chosen to be eigenstates of the other operator in a suitable basis. Thus, when diagonalized accordingly, the Hamiltonian can be split into block sections corresponding to shared eigenvectors of symmetries and the Hamiltonian. In other words, the Hilbert Space $\mathcal{H}$ splits into invariant subspaces that transform irreducibly under G

$$\mathcal{H} = \bigoplus_\lambda \mathcal{H}_\lambda \tag{8}$$

Each $\mathcal{H}_\lambda$ carries an irreducible representation of the group.

Say we have a block diagonal Hamiltonian with a coupled subspace. Then, Eigenstates for that subspace will be linear combinations of states from each subspace

V. GROUP ALGEBRAS AND PROJECTION METHODS

In order to construct a Hamiltonian on our Hilbert space, one good way to proceed is to start generally with all possible Hamiltonian operators and narrow down the possibilities using symmetries. To classify all G -invariant operators on our Hilbert space, we use something known as a *group algebra*, denoted $\mathbb{C}[G]$. Don't worry about what an "algebra" actually is; it's just a name for all we care about. This group algebra $\mathbb{C}[G]$ is defined to be the set of all *complex-linear combinations* of the group operations (hence the insertion of the $\mathbb{C}$ out front)

$$\mathbb{C}[G] := \bigoplus_{\phi \in G} \mathbb{C}[\phi], \quad (9)$$

where $\mathbb{C}[\phi]$ represents all formal multiples $\lambda\phi$, where $\lambda \in \mathbb{C}$.

Another important structure we are concerned with is the *group center* $\mathcal{Z}(\mathbb{C}[G])$ of the group algebra $\mathbb{C}[G]$. This is a subset $\mathcal{Z}(\mathbb{C}[G]) \subseteq \mathbb{C}[G]$ consisting of those elements of the algebra that commute with all elements of G . Thus, these elements represent operations that are fully symmetric (invariant) with respect to G . One trivial, but useful, property we can note is that summations over (polynomials of) group elements $\sum_{\phi \in G} \hat{U}(\phi)$ result in operators still within the center. An important invariant operator, which we can actually find computationally from a symmetry group, is the *symmetrizing operator* (or Reynolds averaging operator)

$$\hat{O}_{\text{sym}} = \frac{1}{|G|} \sum_{\phi \in G} \hat{U}(\phi), \quad (10)$$

which makes another operator or state *symmetric* with respect to the group G . In ??, we used this operator to create crystal group-invariant functions.

A. Symmetrizer? I Hardly Know Her

Returning to the symmetry-invariant subspaces of our Hilbert Space $\mathcal{H}$, we can *project* onto a subspace for an irrep using something known as a *Young Projector* or *Generalized Young Symmetrizers (GYS)*. A complete set of GYS is a set $\{P_j\}_{j=1}^m$ of elements of the group algebra $\mathbb{C}[G]$, which satisfy all the properties we typically require of projections. However, we also impose one additional requirement; that of *primitivity*:

$$P_j \phi P_j = \lambda \phi P_j \quad \text{for all } \phi \in G. \quad (11)$$

This guarantees that P_j will project onto an irreducible subspace, as P_j acts like the identity. Such a set of GYS always exists [From appendix III [24] of [10] of [Clebsch Gordan Decom CITE]] Now, similar to our usual projections, we can use these symmetrizers to write

$$\mathbb{C}[G] = \mathbb{C}[G]\mathbb{I} = \mathbb{C}[G](\sum_j P_j) = \sum_j \mathbb{C}[G]P_j = \sum_{j=1}^q \mathcal{C}_j, \quad (12)$$

where $\mathcal{C}_j := \mathbb{C}[G]P_j$ can be understood as all operators projected onto this symmetry-invariant subspace (why?).

These are very important: Physically, these *isolate the degrees of freedom* that are invariant under G . (why??)

B. Symmetry-Adapted Basis

Assuming we have a complete set of these projection operators $\{P_j\}$, we can use them to extract the invariant subspaces from our Hilbert Space $\mathcal{H}$:

$$\mathcal{H}_\lambda = P_\lambda \mathcal{H} \quad (13)$$

Any symmetric operator $\hat{F}$ that commutes with every element of G , so $\hat{F} \in \mathcal{Z}(\mathbb{C}[G])$, will leave the subspace $P_j \mathcal{H}$ unchanged. Applying Schur's Lemma, we can choose an appropriate basis such that the operator $\hat{F}$ is block-diagonal. We call this basis the *symmetry-adapted basis*. Thus, applying to the Hamiltonian $\hat{H}$ in this basis, we can get its restriction to each invariant subspace $\hat{H}_\lambda = P_\lambda \hat{H} P_\lambda$. Then, our Hamiltonian becomes

$$\hat{H} = \bigoplus_{\lambda} \hat{P}_\lambda \hat{H} \hat{P}_\lambda, \quad (14)$$

which is equivalent to our formulation in ???. Unlike our formulation, however, this does not separate the invariant part of the Hamiltonian from the additional degrees of freedom on the multiplicity space; it considers them together as part of one block.

Each block acts only on one invariant subspace, so dynamics of different blocks do not mix. Note that $\hat{P}_\lambda$ is G -invariant, so $\hat{P}_\lambda \in \mathcal{Z}(\mathbb{C}[G])$

The group algebra $\mathbb{C}[G]$ contains all projections onto symmetric subspaces (GYS) as these obey the symmetries?

When the Hamiltonian is block-diagonal, so it can be written as a direct sum of projection subspaces, its exponential is given by the direct sum of the block-exponential

VI. APPLICATIONS

A. Spontaneous Emission

VII. CONCLUSION

Conclusion

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Appendix A: Extras

1. Invariance

For a symmetry group G , an invariant subspace of our HS $\mathcal{V} \leq \mathcal{H}$ is one such that the *orbit* of every state in the subspace lies completely within the subspace $\mathcal{V}$. This extends the “closure” property we saw with abstract groups. An invariant subspace can be thought of a closed “container” of many orbits: all possible places a state can get to through the transformations.

2. Conservation in the Continuum

Many physical symmetries manifest continuously. For example, isotropic spherical symmetry of the free-particle can be seen as the group of all possible rotations in three-dimensions, denoted⁴ $\text{SO}(3)$.

These symmetries are represented by *Lie groups*, which are groups with a continuum number of elements.

For example, unlike a discrete symmetry, continuous symmetries can be made arbitrarily small, such as a rotation by an infinitesimally small angle ϵ . We can write such an infinitesimal symmetry transformation as the identity plus a linear correction term⁵:

$$\hat{U}(\epsilon) \approx \mathbb{I} - i\epsilon\hat{A}. \quad (\text{A1})$$

We can then build finite transformations by repeated application of infinitesimal ones

$$\hat{U}(t) = \lim_{n \rightarrow \infty} \left(\mathbb{I} - i\frac{t}{n}\hat{A} \right)^n = e^{-it\hat{A}}, \quad \epsilon \mapsto \frac{t}{n}, \quad (\text{A2})$$

⁴The group is termed $\text{SO}(3)$, as it is the “special orthogonal group in 3-dimensions,” reflecting its unitary properties.

⁵The prefactor of $\hat{A}$ should actually be $\frac{i}{\hbar}\epsilon$, but the $\hbar$ is omitted for pedagogical clarity. Also, with the sign of the linear term, we have implicitly adopted the clockwise rotation standard.

where the final equality comes from the realization that the limit expression of applying infinitesimal operations builds the Taylor series expansion of the exponential function in terms of the operator $\hat{A}$. The operator $\hat{A}$ must additionally be self-adjoint⁶, so it is an observable.

As a result of this construction, we call the operator $\hat{A}$ the *generator* of the Lie group⁷.

If a continuous symmetry generated by an observable $\hat{A}$ is present in a system, and thus the Hamiltonian $\hat{H}$ commutes with the unitary operator $\hat{U}(\hat{A}) = e^{-it\hat{A}}$ as $[\hat{U}(\hat{A}), \hat{H}] = 0$, then, we can see from the previous section that $\hat{H}$ also commutes with the observable $\hat{A}$ as well. They are said to be compatible. We can gain something stronger, though. In an isolated quantum system, the evolution is governed by the Heisenberg equation of motion, which says⁹

$$\frac{d}{dt}\langle\hat{A}\rangle = i\langle[\hat{H}, \hat{A}]\rangle. \quad (\text{A3})$$

Thus, when we have a system obeying a continuous symmetry that is generated by an observable $\hat{A}$, the expectation value $\langle\hat{A}\rangle$ is constant in time and thus a *conserved quantity*. This leads us to the desired correspondence ?? and *Noether's Theorem* [4], which we abbreviate:

Theorem 1 (Noether's Theorem). *For every continuous symmetry of a physical system, there exists a corresponding conservation law.*

Appendix B: Proofs

This section consists of proofs for claims made in the main text body that may not have been adequately justified.

Proposition 1 (Commuting operators share eigenvectors). *Let $\hat{H}$ and $\hat{U}$ be normal linear operators such that $[\hat{H}, \hat{U}] = 0$. Then, $\hat{H}$ and $\hat{U}$ can be simultaneously diagonalized. In other words, a basis can be chosen such that they share eigenvectors.*

Proof. Let $\hat{H}$ and $\hat{U}$ be as in the proposition, so $\hat{U}^{-1}\hat{H}\hat{U} = \hat{H}$. Then, let ψ be an eigenvector of $\hat{H}$, so $\hat{H}\psi = \lambda\psi$. Thus,

$$\hat{H}\hat{U}\psi = \hat{U}\hat{H}\psi = \lambda\hat{U}\psi.$$

Therefore, $\hat{U}\psi$ is an eigenvector of $\hat{H}$. If λ is non-degenerate, then its eigenspace is one-dimensional, so we must have $\hat{U}\psi = c\psi$ for some scalar c . In this case, ψ would be an eigenvector of $\hat{U}$. However, if λ has some degeneracy, then $\hat{H}$ has an eigenspace E_λ spanned by eigenvectors with eigenvalue λ . Thus, $\hat{U}\psi \in E_\lambda$. Then, the eigenspace E_λ is invariant under $\hat{U}$ as $\psi \in E_\lambda$ was arbitrary. As $\hat{U}$ is normal, it can be diagonalized in this subspace, so we can choose a new basis of E_λ such that this basis consists of eigenvectors of both $\hat{U}$ and $\hat{H}$. $\square$

Proposition 2. *Let $\hat{U}$ be a unitary operator. Then, it has eigenvalues on the complex unit circle:*

$$\sigma(\hat{U}) \subseteq \{e^{i\theta} : \theta \in [0, 2\pi)\}. \quad (\text{B1})$$

Proof. Let $\hat{U}$ be a unitary operator. It is a standard result from linear algebra that, for an invertible operator $\hat{U}$, the eigenvalues of $\hat{U}^{-1}$ are the inverses of those of $\hat{U}$. However, the eigenvalues of the adjoint of an operator $\hat{U}^*$ are the complex conjugate of those of $\hat{U}$. Since $\hat{U}$ is unitary, we have $\hat{U}^* = \hat{U}^{-1}$, and so $\lambda^* = \lambda^{-1}$ for all $\lambda \in \sigma(\hat{U}) \subseteq \mathbb{C}$. This is exactly the complex unit circle: $\|\lambda\|^2 = \lambda^*\lambda = \lambda^{-1}\lambda = 1$. $\square$

⁶Reasoning beyond the scope of this paper, but see Stone's Theorem [3], which states that to every strongly continuous one-parameter unitary group, there corresponds a unique self-adjoint operator, and thus an observable (follows from that we need $\hat{U}^* = \hat{U}^{-1}$). Note that this is a key detail behind Noether's Theorem!

⁷A Lie group is not limited to having one generator; it can have multiple, where each one corresponds to a direction in which the system can change, or a degree of freedom in which it has continuous symmetry. For example, $\text{SO}(3)$ needs three generators; one to specify the rotation angle and two more to specify the axis of rotation. This collection of generators forms a vector space spanned by the generators⁸, which is a structure called a *Lie algebra*. It is essentially all the terms that look like Eq. A1.

⁹Assuming the operator $\hat{A}$ has no time dependence: $\partial_t \hat{A} = 0$. And we have again absorbed the factor of $\hbar^{-1}$ on the right-hand side for illustrative purposes.